

SFC increases with Mach number. Thrust initially drops with increasing Mach number and then rises when the ram effect overcomes the increase of momentum drag; the fuel flow, however, will steadily increase with Mach number because of the increased airflow due to the higher stagnation pressure at entry to the compressor, and this effect predominates.

9.6 Methods of displacing the equilibrium running line

It was stated earlier that if the equilibrium running line intersects the surge line it will not be possible to bring the engine up to full power without taking some remedial action. As will be shown in the next chapter, even when clear of the surge line, if the running line approaches it too closely the compressor may surge when the engine is accelerated rapidly.

With most modern compressors, surge is likely to be encountered at low values of $N/\sqrt{T_{01}}$ and is less of a problem at high rotational speeds. Many high-performance axial compressors exhibit a kink in the surge line as shown in Fig. 9.19. A running line, intersecting the surge line at low speeds and at the kink, is also shown. To overcome the problem it is necessary to lower the running line locally in dangerous regions of operation.†

One common method of achieving this is *blow-off*, where air is bled from some intermediate stage of the compressor. Blow-off clearly involves a waste of turbine work, so that the blow-off valve must be designed to operate only over the essential part of the running range. Furthermore, it may be difficult to discharge the air when

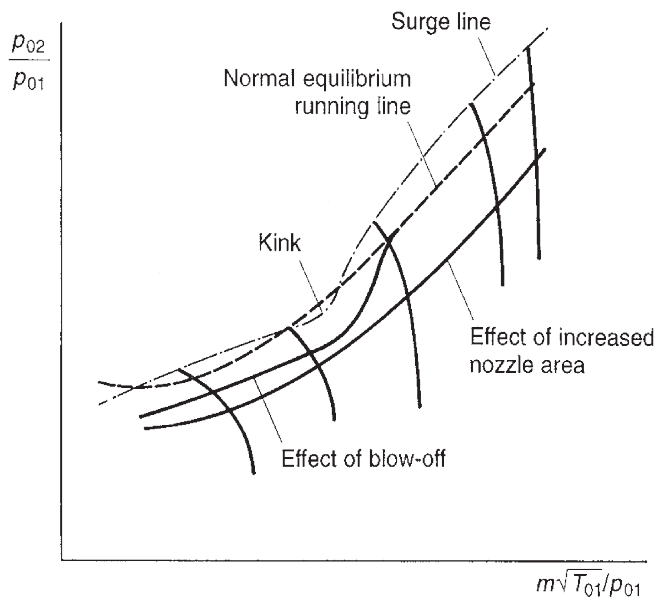


FIG. 9.19 Effect of blow-off and increased nozzle area

† An alternative method is to raise the surge line using variable stators in the compressor.

space around the engine is at a premium as in a nacelle. As an alternative to blow-off a *variable area propelling nozzle* could be used for a jet engine. It will be shown in the next chapter that variable area propelling nozzles serve other useful purposes also.

It can be deduced that either of these methods will produce a reduction in pressure ratio at a given compressor speed and hence will lower the running line. This is most easily demonstrated if we consider operation at the high-speed end of the running range, where the constant $N/\sqrt{T_{01}}$ lines on the compressor characteristic are almost vertical (i.e. $m\sqrt{T_{01}}/p_{01} \approx \text{constant}$) and where both the nozzle and turbine are choked.

If we consider the use of a variable area nozzle first, Fig. 9.20 shows that increasing the nozzle area will cause an increase in turbine pressure ratio and hence non-dimensional temperature drop $\Delta T_{034}/T_{03}$. From equation (9.2), satisfying compatibility of flow,

$$\frac{p_{02}}{p_{01}} = \frac{m\sqrt{T_{01}}}{p_{01}} \times \frac{p_{02}}{p_{03}} \times \sqrt{\frac{T_{03}}{T_{01}}} \times \frac{p_{03}}{m\sqrt{T_{03}}} = K_1 \sqrt{\frac{T_{03}}{T_{01}}} \quad (9.23)$$

where K_1 is a constant because under the assumed conditions the values of $m\sqrt{T_{01}}/p_{01}$, p_{02}/p_{03} and $m\sqrt{T_{03}}/p_{03}$ are all fixed. From equation (9.6), which satisfies compatibility of work,

$$\frac{T_{03}}{T_{01}} = \frac{\Delta T_{012}}{T_{01}} \times \frac{T_{03}}{\Delta T_{034}} \times \frac{c_{pa}}{c_{pg}\eta_m}$$

Now, at a fixed value of compressor speed N the compressor temperature rise is approximately constant; Fig. 9.21 shows a typical compressor characteristic replotted on a basis of temperature rise versus pressure ratio. The wide range of pressure ratio obtained at almost constant temperature rise is due to the significant variation of efficiency when operating at conditions away from the design point. Thus if we assume $\Delta T_{012}/T_{01}$ is constant,

$$\frac{T_{03}}{T_{01}} = \frac{K_2}{(\Delta T_{034}/T_{03})} \quad (9.24)$$

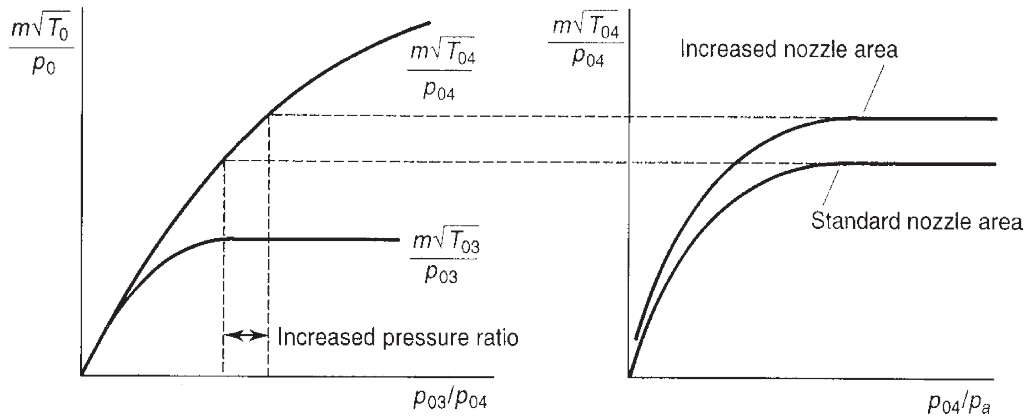


FIG. 9.20 Effect of variable area propelling nozzle

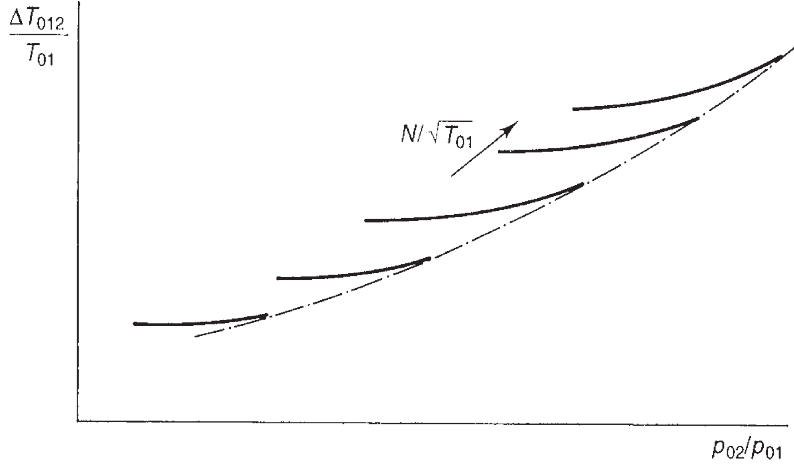


FIG. 9.21 Compressor characteristics

where K_2 is another constant. Combining equations (9.23) and (9.24) gives

$$\frac{p_{02}}{p_{01}} = \frac{K_3}{\sqrt{(\Delta T_{034}/T_{03})}} \quad (9.25)$$

where K_3 is a constant, obtained from K_1 and K_2 .

If the engine speed $N/\sqrt{T_{01}}$ is held constant and the nozzle area is increased, $\Delta T_{034}/T_{03}$ will be increased and equation (9.25) shows that p_{02}/p_{01} will be decreased, hence the running line will be moved away from the surge line. In order to maintain the speed at the required level, it is necessary to reduce the fuel flow; opening the nozzle without reducing the fuel flow would cause the engine to accelerate to a higher speed, as discussed in Chapter 10. It should also be realized that decreasing the nozzle area will move the operating line towards the surge line; reasons for doing this will be discussed in the next chapter.

When considering the effect of bleed, m_1 is no longer equal to m_3 and equation (9.23) is modified to

$$\frac{p_{02}}{p_{01}} = K_1 \sqrt{\frac{T_{03}}{T_{01}}} \times \frac{m_3}{m_1} \quad (9.26)$$

With the nozzle, or power turbine in the case of a shaft power unit, remaining choked, the turbine operating point remains unchanged and equation (9.26) leads to

$$\frac{T_{03}}{T_{01}} = \frac{m_1}{m_3} \times \frac{\Delta T_{012}}{T_{01}} \times \frac{T_{03}}{\Delta T_{034}} \times \frac{c_{pa}}{c_{pg}\eta_m} = K_4 \frac{m_1}{m_3} \quad (9.27)$$

where K_4 is a constant, assuming as before that $\Delta T_{012}/T_{01}$ is constant. Combining equations (9.26) and (9.27) gives

$$\frac{p_{02}}{p_{01}} = K_5 \sqrt{\frac{m_3}{m_1}} \quad (9.28)$$

where K_5 is a constant.

With the use of bleed, m_3 is always less than m_1 , and the result will once again be to reduce the pressure ratio and lower the operating line. Equation (9.27) also shows that the use of bleed will increase the turbine inlet temperature. The reduction in turbine mass flow necessitates an increase in temperature drop to provide the compressor work, and with the non-dimensional temperature drop fixed this implies an increase in turbine inlet temperature. The effects of both bleed and increase in nozzle area can be interpreted physically in terms of a decrease in the restriction to the flow, permitting the compressor to operate at a lower pressure ratio for any given rotational speed.

9.7 Incorporation of variable pressure losses

The size of the ducting for components such as the intake and exhaust of a gas turbine will be determined primarily by the requirement for low pressure losses at maximum power, which will normally correspond to maximum mass flow. The designer must compromise between low pressure losses and small duct sizes, and the choice will be largely determined by the application envisaged. It can be shown by cycle calculations that the effect of a given pressure loss is dependent on the pressure level at which it occurs, and it is essential to keep the terms $\Delta p_0/p_0$ at all stations as low as possible. With pressure losses in the intake and exhaust of stationary gas turbines occurring at essentially atmospheric pressure, it will be realized that these losses are much more critical than pressure losses in the combustion chamber or the air side of a heat-exchanger where the air is entering at the compressor delivery pressure. In a typical installation a pressure loss of 2.5 cm H₂O in the intake may cause about 1 per cent reduction in power output.

Having decided on the size of ducting required to give the desired pressure loss at the design point, the variation of loss with changing operating conditions can be predicted for incorporation in the off-design calculation procedures outlined in this chapter. The velocities in such components as the intake and exhaust ducts will be sufficiently low for the flow to be treated as incompressible, so that Δp_0 will be proportional to the inlet dynamic head. Expressed in non-dimensional terms this implies, as shown by equation (6.2), that

$$\frac{\Delta p_0}{p_0} \propto \left(\frac{m\sqrt{T_0}}{p_0} \right)^2$$

Assuming the appropriate values at the design condition to be $(\Delta p_0/p_0)_D$ and $(m\sqrt{T_0}/p_0)_D$, then at any off-design condition

$$\frac{\Delta p_0}{p_0} = \left(\frac{\Delta p_0}{p_0} \right)_D \left[\left(\frac{m\sqrt{T_0}}{p_0} \right) / \left(\frac{m\sqrt{T_0}}{p_0} \right)_D \right]^2 \quad (9.29)$$

Values of $m\sqrt{T_0}/p_0$ for each component are obtained in the course of the off-design calculations. The pressure loss from equation (9.29) can be incorporated in the

same manner as was the combustion pressure loss $\Delta p_b/p_{02}$, i.e. by making use of such equations as

$$\frac{p_{03}}{p_{02}} = 1 - \frac{\Delta p_b}{p_{02}}$$

Now, however, the pressure ratio across the component is not constant but changes with $m\sqrt{T_0}/p_0$.

In the particular case of the combustion chamber, the more accurate expression of equation (6.1) can be used. This includes the fundamental loss due to heat addition. Combining (6.1) and (6.2), we would have

$$\frac{\Delta p_b}{p_{02}} \propto \left(\frac{m\sqrt{T_{02}}}{p_{02}} \right)^2 \left[K_1 + K_2 \left(\frac{T_{03}}{T_{02}} - 1 \right) \right]$$

Such refinement would not normally be worthwhile because the fundamental loss is a small proportion of the total.

9.8 Power extraction

Gas turbines must provide power to drive auxiliaries such as fuel pumps, oil pumps and hydraulic pumps. The power is normally extracted from the gas generator in a free turbine or jet engine and from the main rotor in a single-shaft engine. In the case of an industrial engine the accessories are often mounted on a skid beside the engine, which simplifies maintenance activities. On an aircraft engine, however, the accessories must all be mounted on the engine with particular attention paid to minimizing frontal area.

Power extraction can be handled by modifying the work compatibility equation so that

$$\text{turbine power} = \text{compressor power} + \text{auxiliary power}$$

The effect of power extraction, then, is to increase the turbine output at a given compressor speed, requiring an increase in fuel flow and consequently turbine inlet temperature, moving the operating point towards surge. In the aircraft application, the engine must also provide power for aircraft accessories such as generators and hydraulic pumps; it should be noted that these power requirements may remain at a high level when the engine is operating at low power, such as during the descent. Thus the operating line is displaced towards surge over the whole running range, and power extraction must always be considered in establishing surge margin. Figure 9.22 shows a typical aero-engine gearbox. The radial drive-shaft will be driven off the main shaft by bevel gears, and will be used to transmit power for engine starting; an external gearbox is shown, and it can be observed that it is crescent shaped to permit mounting on an external casing.

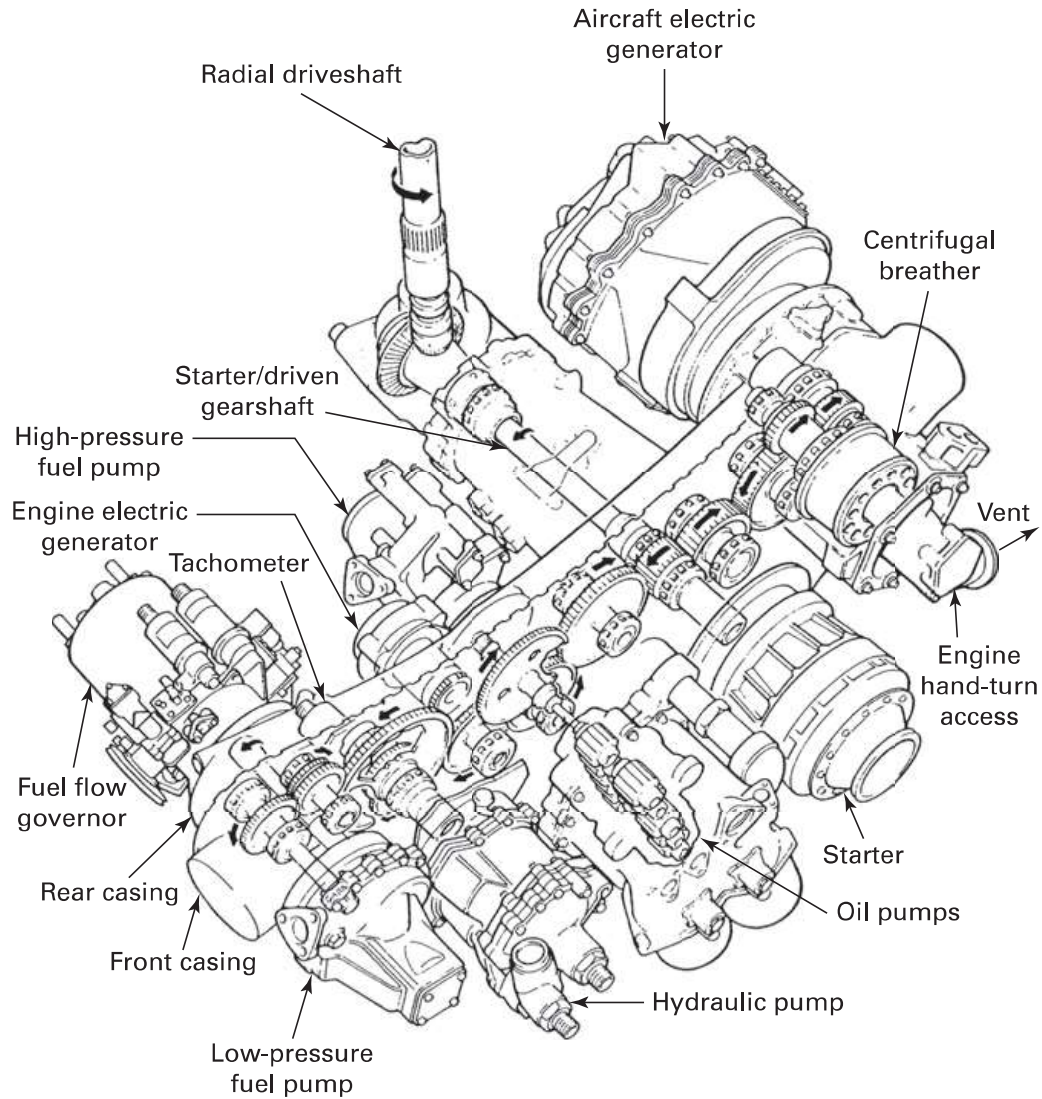


FIG. 9.22 An external gearbox and accessory units [courtesy Rolls-Royce plc]